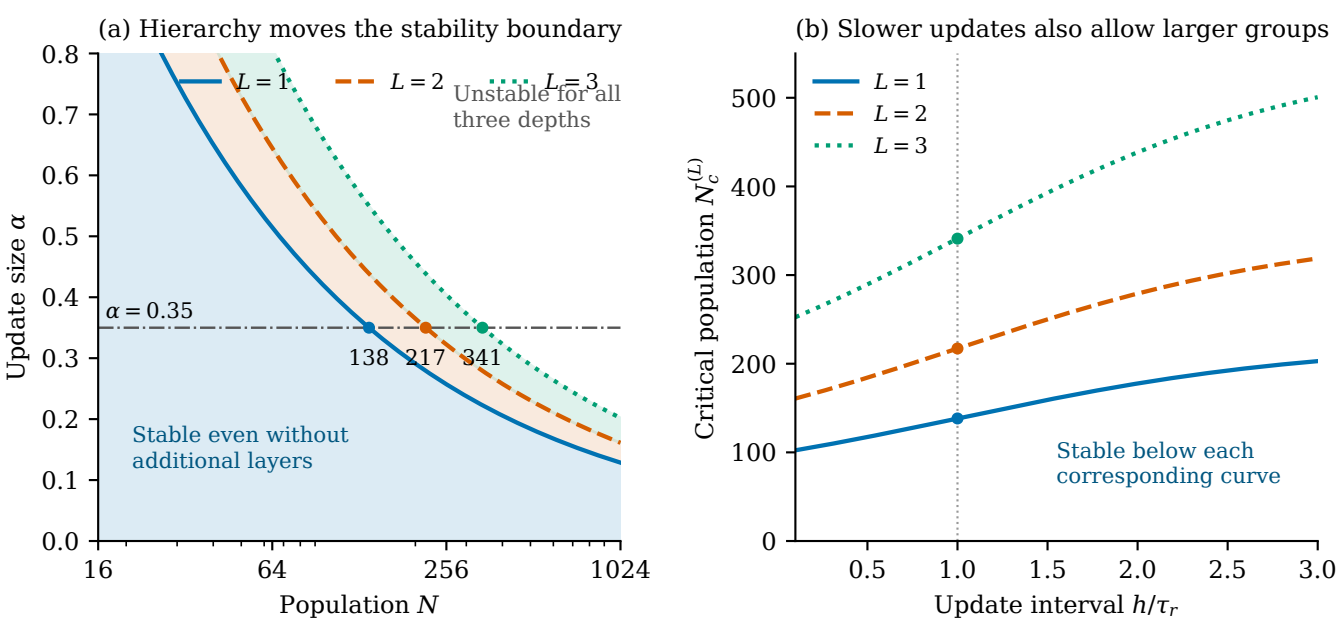
(a) Hierarchy moves the stability boundary
L = 1
L = 2
L = 3
Unstable for all three depths
α = 0.35
138
217
341
Stable even without additional layers
Update size α
Population N
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
16
64
256
1024
(b) Slower updates also allow larger groups
L = 1
L = 2
L = 3
Stable below each corresponding curve
Critical population $N_c^{(L)}$
Update interval h/τ_r
500
400
300
200
100
0
0.5
1.0
1.5
2.0
2.5
3.0